CHAPTER 8

DATABASE MANAGEMENT SYSTEM

CHAPTER ESSENCE AT A GLANCE:
Understand database management principles, 3-tier ANSI-SPARC data abstraction, relational schema models, entity-relationship diagrams, and relational keys.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Overcoming File System Deficiencies:** Database management systems centralize organizational data storage, eliminating duplicate redundancy, resolving data inconsistencies, enforcing integrity constraints, and providing robust multi-user security.

• **Three-Tier Data Abstraction:** ANSI architecture separates user perception into three distinct levels: the physical storage level detailing how bits are stored, the logical conceptual level defining what data is stored, and the external view level presenting customized perspectives.

• **Relational Terminology and Dimensions:** The relational model represents information as two-dimensional tables termed relations, where each row forms an individual record tuple, and each column represents a typed attribute; the count of columns defines the degree, and the count of rows defines cardinality.

• **Key Constraints and Identity:** A candidate key represents a minimal superkey that uniquely identifies tuples; one candidate key is selected as the primary key that strictly forbids null entries, while remaining candidates become alternate keys.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions

$$\text{External View Level} \longrightarrow \text{Conceptual Logical Level} \longrightarrow \text{Internal Physical Level}$$

ANSI-SPARC three-level database abstraction architecture

$$\text{Super Key} \supseteq \text{Candidate Key} \xrightarrow{\text{Chosen}} \text{Primary Key}$$

Database integrity key selection and containment hierarchy

$$\text{Selectivity Ratio} = \frac{\text{Selected Tuples } (\sigma)}{\text{Total Relation Tuples } (R)}$$

Relational algebra selection predicate selectivity ratio

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Primary keys must enforce both entity integrity (strictly non-null) and absolute uniqueness across all relation tuples.

• **Alert:** Foreign keys establish referential integrity by referencing valid primary key values in a parent relation or remaining null.

• **Alert:** The degree of a relation is its column or attribute count, whereas cardinality represents the total row or tuple count.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the limitations of traditional file processing systems and how DBMS overcomes them. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (File Processing Limitations):** 1. Data redundancy (duplicate copies). 2. Data inconsistency (mismatched values). 3. Difficulty in accessing data. 4. Data isolation (scattered in different file formats). 5. Integrity problems. 6. Atomicity problems. 7. Concurrent access anomalies. 8. Security problems. [3 Marks]

• **Key Exam Result:** DBMS eliminates redundancy, inconsistency, and concurrency conflicts

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions

2. Explain the three-level database architecture (Physical, Conceptual, and External or View level). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Three Levels of Abstraction):** 1. Physical or Internal Level: Lowest level; describes how data is actually stored on hardware disks (blocks, indexing, B-trees). 2. Conceptual or Logical Level: Middle level; describes what data is stored and relationships among data (tables, schemas, constraints). 3. View or External Level: Highest level; describes user-specific views, hiding irrelevant data. [3 Marks]

• **Key Exam Result:** Physical (how stored) --> Conceptual (what stored) --> View (user views)

$$\text{External View Level} \longrightarrow \text{Conceptual Logical Level} \longrightarrow \text{Internal Physical Level}$$

ANSI-SPARC three-level database abstraction architecture

3. Define Data Redundancy, Data Inconsistency, and Data Independence in DBMS. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Definitions):** (a) Data Redundancy: Unnecessary duplication of the same data across multiple files. (b) Data Inconsistency: Different copies of the same data contain conflicting values due to partial updates. (c) Data Independence: Ability to modify schema at one level without altering schema at next higher level (Logical and Physical Data Independence). [3 Marks]

• **Key Exam Result:** Redundancy (duplication), Inconsistency (conflicts), Independence (schema decoupling)

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions

4. Explain key relational model terms: Relation, Tuple, Attribute, Domain, Degree, and Cardinality. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Relational Model Terms):** Proposed by E.F. Codd; data represented as 2D tables (Relations). Relation: Table with rows and columns. Tuple: A row in a relation (represents a single record). Attribute: A column in a relation (represents a property or field). Degree: Total number of attributes (columns). Cardinality: Total number of tuples (rows). Domain: Set of permitted values for an attribute. [3 Marks]

• **Key Exam Result:** Relation (table), Tuple (row), Attribute (col), Degree (#cols), Cardinality (#rows)

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions

5. Explain the different types of keys in relational databases: Primary Key, Candidate Key, and Alternate Key. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Keys Explanation):** 1. Candidate Key: A minimal set of attributes that uniquely identifies each tuple in a relation. 2. Primary Key: The specific candidate key chosen by database designer to uniquely identify tuples (cannot be NULL). 3. Alternate Key: Candidate keys that were not chosen as the primary key. 4. Foreign Key: An attribute in a table that references the primary key of another table, establishing a relationship. **[4 Marks]**

• **Key Exam Result:** Primary (chosen unique identifier), Candidate (all uniques), Foreign (referencing key)

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions

6. Explain the role and key responsibilities of a Database Administrator (DBA). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Role of DBA):** A Database Administrator is the central person responsible for total management, performance, and security of database system: 1. Schema definition and storage structure. 2. Granting user authorizations and permissions. 3. Routine backups and disaster recovery. 4. Performance tuning and monitoring. **[2 Marks]**

• **Key Exam Result:** DBA manages schema, user access security, backups, and performance

$$\text{Super Key} \supseteq \text{Candidate Key} \xrightarrow{\text{Chosen}} \text{Primary Key}$$

Database integrity key selection and containment hierarchy

7. Differentiate between a Strong Entity Set and a Weak Entity Set with an example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Strong Entity Set):** Has sufficient attributes to form a primary key on its own; exists independently (represented by single rectangle). **[1 Mark]**

• **Step 2 (Weak Entity Set):** Does not have a primary key on its own; depends on existence of an identifying strong entity set; has partial key or discriminator (represented by double rectangle). E.g. Dependent of Employee. **[1 Mark]**

• **Key Exam Result:** Strong (has primary key) vs Weak (double rectangle, requires identifying parent)

$$\text{Degree} = \text{Column Count} \quad | \quad \text{Cardinality} = \text{Row Count}$$

Fundamental relational table structural dimensions